THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 13: **Our Environment** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In the food chain Phytoplankton, Zooplankton, Small fish, Large fish the secondary consumer is 1

(a) small fish **(b)** large fish **(c)** zooplankton **(d)** phytoplankton

2. In a food chain, DDT concentration is highest in 1

(a) zooplankton **(b)** small fish **(c)** eagle **(d)** phytoplankton

3. Segregation of waste means 1

(a) dumping in sea
(b) burning
(c) separating biodegradable from non-biodegradable
(d) burying

4. Which of the following is the correct food chain? 1

(a) Hawk, bird, caterpillar, plant **(b)** Bird, plant, hawk
(c) Plant, caterpillar, bird, hawk **(d)** Plant, hawk, bird

5. Which of the following would show the greatest biomagnification? 1

(a) A hawk that eats snakes **(b)** A snail
(c) Grass **(d)** A grasshopper

6. A sewage treatment plant converts waste into 1

(a) fuel only **(b)** plastic
(c) manure and clean water **(d)** glass

7. If the producers of a food chain have 50,000 J of energy, the tertiary consumers receive 1

(a) 5000 J **(b)** 500 J **(c)** 50 J **(d)** 5 J

8. Which one is a harmful effect of CFCs? 1

(a) Increasing oxygen **(b)** Cooling the sun
(c) Making plants grow **(d)** Destroying the ozone layer

9. Composting is the process of 1

(a) burning plastic **(b)** recycling glass
(c) converting vegetable waste into manure **(d)** melting metals

10. If a top carnivore receives 10 J of energy through 3 transfers from producers, the energy at the producers is 1

(a) 1000 J **(b)** 10,000 J **(c)** 100,000 J **(d)** 100 J

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A food chain has the organisms: grass, frog, grasshopper, snake. Arrange them in the correct order and identify the tertiary consumer. 2

12. How can a food web be more stable than a food chain? 2

13. Why is DDT concentration in humans higher than in plants? 2

14. If a lake has producers with 20,000 J, find the energy available to the primary and the secondary consumer. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In a food chain, Grass, Deer, Tiger, find the energy available to the deer and to the tiger if grass has 12,000 J. State the law used. 3

16. A farmer sprays pesticide on a crop. Describe how it reaches the top consumers. 3

17. Why is there no use of 'dead end' in a food chain of an ecosystem which has decomposers? 3

OR

Why do we get less energy by eating meat than by eating plants? 3

18. Why are the plastics a great threat to the environment although they are used widely? 3

OR

Explain why the number of trophic levels is limited to four or five. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Construct a food chain of a pond with five organisms and calculate the energy available at each level if the producers have 100,000 J. Which level has the least energy? 5

OR

Explain the flow of energy in an ecosystem. Why can the energy not be recycled? 5

20. Explain why the number of individuals decreases at each successive trophic level and why the energy transfer is only 10 per cent. 5

OR

Explain the roles of producers, herbivores, carnivores and decomposers in the ecosystem and how the removal of any one of them can affect the others. 5

END OF QUESTION PAPER — Total: 40 Marks